UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

STATISTICAL SIGNAL PROCESSING - *Prova scritta del 29/06/2023*

EXERCISE

It is given an Auto-Regressive (AR) process $S[n]$ of order 1, $S[n] = \rho S[n-1] + I[n]$, where $I[n]$ is the zero-mean innovation process with variance σ_I^2.

(1) Derive the system function $L(z)$ of the innovation filter, its zeros and poles, and determine when it is a low-pass, high-pass, or band-pass filter, according to the value of ρ. Derive the power of the process σ_S^2, the autocorrelation function (ACF) $R_S[m] = E\{S[n]S[n+m]\}$ and the power spectral density (PSD) $S_S\left(e^{j2\pi f}\right)$ of the process $S[n]$ and plot them for $\rho = 0.95$ and $\sigma_I^2 = 2$.

(2) Derive the linear minimum mean square error (LMMSE) estimator of the process at time *n*+1, $S[n+1]$, given the observed data $S[n]$ and $S[n-1]$. Derive its mean square error (MSE) $\sigma_\varepsilon^2 \triangleq E\{\varepsilon^2[n+1]\} = E\left\{\left(S[n+1] - \hat{S}[n+1]\right)^2\right\}$. Calculate numerically the expression of the estimator and its MSE when $\rho = 0.95$ and $\sigma_I^2 = 2$.

(3) Assuming that the signal of interest (SoI) $S[n]$ is observed in the presence of additive white Gaussian noise (AWGN), i.e. $X[n] = S[n] + W[n]$, where $W[n]$ is AWGN with variance σ_W^2, derive the LMMSE estimator of $S[n+1]$, given the observed data given the observed data $X[n]$ and $X[n-1]$ as a function of ρ and of the *Signal to Noise Ratio* (SNR) $\gamma = \sigma_S^2 / \sigma_W^2$. Derive its MSE. Investigate what happens to $\hat{S}[n+1]$ when $\rho \to 0$ and $\rho \to 1$; explain the results. Calculate numerically the expression of $\hat{S}[n+1]$ and its MSE when $\rho = 0.95$, $\sigma_I^2 = 2$, and $\sigma_W^2 = 1$; compare the MSE with the one previously derived.

Solution.

(1) $S[n]=\rho S[n-1]+I[n]$

$$S(z)=\rho z^{-1}S(z)+I(z) \quad \Rightarrow \quad L(z)=\frac{S(z)}{I(z)}=\frac{1}{1-\rho z^{-1}}=\frac{z}{z-\rho}$$

The innovation filter has 1 zero at $z=0$ and 1 pole at $z=\rho$. Hence, the innovation filter is a high-pass filter if $\rho<0$ and a low-pass filter if $\rho>0$, and so the process.

The ACF and the PSD are given by (see the course vugraphs for the derivation):

$R_S[m]=E\{S[n]S[n+m]\}=\frac{\sigma_I^2}{1-\rho^2}\rho^{|m|}$, hence $\sigma_S^2=R_S[0]=\frac{\sigma_I^2}{1-\rho^2}$.

$$S_S\left(e^{j2\pi f}\right)=\sigma_I^2\left|L\left(e^{j2\pi f}\right)\right|^2=\sigma_I^2\left|\frac{1}{1-\rho e^{-j2\pi f}}\right|^2=\frac{\sigma_I^2}{1+\rho^2-2\rho\cos(2\pi f)}$$

When $\rho=0.95$, $S[n]$ is a low-pass process with narrow bandwidth.

(2) $\hat{S}[n+1]=a_1S[n]+a_2S[n-1]+b$

Since the signal process is zero mean, we have:

$$E\{\varepsilon[n+1]\}=E\{S[n+1]-a_1S[n]-a_2S[n-1]-b\}=-b=0 \quad \Rightarrow \quad b=0.$$

$$E\{\varepsilon[n+1]S[n-m]\}=0, \quad m=0,1$$
$$E\{\left(S[n+1]-\hat{S}[n+1]\right)S[n-m]\}=0, \quad m=0,1$$
$$E\{\left(S[n+1]-a_1S[n]-a_2S[n-1]\right)S[n-m]\}=0, \quad m=0,1$$
$$R_S[m+1]-a_1R_S[m]-a_2R_S[m-1]=0, \quad m=0,1$$

$$\begin{cases} R_S[1]-a_1R_S[0]-a_2R_S[-1]=0 \\ R_S[2]-a_1R_S[1]-a_2R_S[0]=0 \end{cases}$$

$$\begin{bmatrix} R_S[0] & R_S[-1] \\ R_S[1] & R_S[0] \end{bmatrix}\begin{bmatrix} a_1 \\ a_2 \end{bmatrix}=\begin{bmatrix} R_S[1] \\ R_S[2] \end{bmatrix}$$

$$R_S[0]=\frac{\sigma_I^2}{1-\rho^2},\quad R_S[1]=R_S[-1]=\frac{\sigma_I^2}{1-\rho^2}\rho,\quad R_S[2]=\frac{\sigma_I^2}{1-\rho^2}\rho^2$$

$$\frac{\sigma_I^2}{1-\rho^2}\begin{bmatrix}1 & \rho\\ \rho & 1\end{bmatrix}\begin{bmatrix}a_1\\ a_2\end{bmatrix}=\frac{\sigma_I^2}{1-\rho^2}\begin{bmatrix}\rho\\ \rho^2\end{bmatrix}$$

$$\begin{bmatrix}a_1\\ a_2\end{bmatrix}=\begin{bmatrix}1 & \rho\\ \rho & 1\end{bmatrix}^{-1}\begin{bmatrix}\rho\\ \rho^2\end{bmatrix}=\frac{\rho}{1-\rho^2}\begin{bmatrix}1 & -\rho\\ -\rho & 1\end{bmatrix}\begin{bmatrix}1\\ \rho\end{bmatrix}=\begin{bmatrix}\rho\\ 0\end{bmatrix}$$

$$\hat{S}[n+1]=\rho S[n]$$

$$\begin{aligned}\sigma_\varepsilon^2&=E\{\varepsilon^2[n+1]\}=E\left\{\left(S[n+1]-\hat{S}[n+1]\right)S[n+1]\right\}\\ &=E\{(S[n+1]-\rho S[n])S[n+1]\}\\ &=R_S[0]-\rho R_S[1]=\frac{\sigma_I^2}{1-\rho^2}-\rho\cdot\frac{\sigma_I^2}{1-\rho^2}\rho=\sigma_I^2\end{aligned}$$

When $\rho=0.95$ and $\sigma_I^2=2$:

$$\begin{bmatrix}a_1\\ a_2\end{bmatrix}=\begin{bmatrix}\rho\\ 0\end{bmatrix}=\begin{bmatrix}0.95\\ 0\end{bmatrix},\quad \hat{S}[n+1]=0.95S[n],\quad MSE=\sigma_\varepsilon^2=\sigma_I^2=2$$

(3) $\hat{S}[n+1]=a_1X[n]+a_2X[n-1]+b$

Since $S[n]$ and $X[n]=S[n]+W[n]$ are zero mean processes we have:

$E\{\varepsilon[n+1]\}=0 \quad\Rightarrow\quad b=0$.

$$E\{\varepsilon[n+1]X[n-m]\}=0,\quad m=0,1$$

$$E\left\{\left(S[n+1]-\hat{S}[n+1]\right)X[n-m]\right\}=0,\quad m=0,1$$

$$E\{(S[n+1]-a_1X[n]-a_2X[n-1])X[n-m]\}=0,\quad m=0,1$$

$$R_{SX}[-(m+1)]-a_1R_X[m]-a_2R_X[m-1]=0,\quad m=0,1$$

where:

$$R_{SX}[m]=E\{S[n]X[n+m]\}=E\{S[n](S[n+m]+W[n+m])\}=E\{S[n]S[n+m]\}=R_S[m]=\frac{\sigma_I^2}{1-\rho^2}\rho^{|m|}$$

$$R_X[m]=E\{X[n]X[n+m]\}=R_S[m]+R_W[m]=\frac{\sigma_I^2}{1-\rho^2}\rho^{|m|}+\sigma_W^2\delta[m]$$

$$\begin{cases} a_1 R_X[0] + a_2 R_X[-1] = R_{SX}[-1] = R_S[1] \\ a_1 R_X[1] + a_2 R_X[0] = R_{SX}[-2] = R_S[2] \end{cases}$$

$$\begin{bmatrix} R_X[0] & R_X[-1] \\ R_X[1] & R_X[0] \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} R_S[1] \\ R_S[2] \end{bmatrix}$$

$$R_X[m] = \frac{\sigma_I^2}{1-\rho^2}\rho^{|m|} + \sigma_W^2 \delta[m], \quad \gamma = \frac{\sigma_S^2}{\sigma_W^2} = \frac{\sigma_I^2}{(1-\rho^2)\sigma_W^2}$$

$$R_X[0] = \frac{\sigma_I^2}{1-\rho^2} + \sigma_W^2 = \frac{\sigma_I^2}{1-\rho^2}\left(1+\frac{1}{\gamma}\right), \quad R_X[1] = R_S[1] = \frac{\sigma_I^2}{1-\rho^2}\rho, \quad R_S[2] = \frac{\sigma_I^2}{1-\rho^2}\rho^2$$

$$\frac{\sigma_I^2}{1-\rho^2}\begin{bmatrix} 1+\frac{1}{\gamma} & \rho \\ \rho & 1+\frac{1}{\gamma} \end{bmatrix}\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \frac{\sigma_I^2}{1-\rho^2}\begin{bmatrix} \rho \\ \rho^2 \end{bmatrix}$$

$$\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 1+\frac{1}{\gamma} & \rho \\ \rho & 1+\frac{1}{\gamma} \end{bmatrix}^{-1}\begin{bmatrix} \rho \\ \rho^2 \end{bmatrix} = \frac{\rho}{\left(1+\frac{1}{\gamma}\right)^2 - \rho^2}\begin{bmatrix} 1+\frac{1}{\gamma} & -\rho \\ -\rho & 1+\frac{1}{\gamma} \end{bmatrix}\begin{bmatrix} 1 \\ \rho \end{bmatrix}$$

$$\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \frac{\rho}{\left(1+\frac{1}{\gamma}\right)^2 - \rho^2}\begin{bmatrix} 1+\frac{1}{\gamma}-\rho^2 \\ \frac{1}{\gamma}\rho \end{bmatrix} \quad \Rightarrow \quad a_1 = \frac{\rho\left(1+\frac{1}{\gamma}-\rho^2\right)}{\left(1+\frac{1}{\gamma}\right)^2 - \rho^2}, \quad a_2 = \frac{\frac{1}{\gamma}\rho^2}{\left(1+\frac{1}{\gamma}\right)^2 - \rho^2}$$

$$\hat{S}[n+1] = \frac{\rho}{\left(1+\frac{1}{\gamma}\right)^2 - \rho^2}\left[\left(1+\frac{1}{\gamma}-\rho^2\right)X[n] + \frac{\rho}{\gamma}X[n-1]\right]$$

$$\begin{aligned}\sigma_\varepsilon^2 &= E\left\{\varepsilon^2[n+1]\right\} = E\left\{\left(S[n+1]-\hat{S}[n+1]\right)S[n+1]\right\}\\ &= E\left\{\left(S[n+1]-a_1X[n]-a_2X[n-1]\right)S[n+1]\right\}\\ &= R_S[0]-a_1R_{SX}[-1]-a_2R_{SX}[-2] = R_S[0]-a_1R_S[1]-a_2R_S[2]\\ &= \frac{\sigma_I^2}{1-\rho^2}\left(1-\rho a_1-\rho^2 a_2\right)\end{aligned}$$

Case $\rho \to 0$:

$$\lim_{\rho\to 0} a_1 = 0, \quad \lim_{\rho\to 0} a_2 = 0 \quad \Rightarrow \quad \hat{S}[n+1]=0$$

In fact in this case the observed data are uncorrelated with what we want to estimate, hence the LMMSE estimator is the a priori mean of $S[n+1]$, which is 0.

Case $\rho \to 1$:

$$\lim_{\rho\to 1} a_1 = \frac{\frac{1}{\gamma}}{\left(1+\frac{1}{\gamma}\right)^2-1} = \frac{\frac{1}{\gamma}}{\frac{2}{\gamma}+\frac{1}{\gamma^2}} = \frac{\gamma}{2\gamma+1}, \quad \lim_{\rho\to -1} a_2 = \frac{\frac{1}{\gamma}}{\left(1+\frac{1}{\gamma}\right)^2-1} = \frac{\gamma}{2\gamma+1}$$

$$\Rightarrow \quad \hat{S}[n+1] = \frac{2\gamma}{2\gamma+1}\left(\frac{X[n]+X[n-1]}{2}\right)$$

In fact, in this case the signal samples are fully correlated, so the correlation between $S[n+1]$ and $X[n]$ is the same as the correlation between $S[n+1]$ and $X[n-1]$, hence they have the same weight.

When $\rho = 0.95$, $\sigma_I^2 = 2$, and $\sigma_W^2 = 1$:

$$\gamma = 20.5128, \quad \hat{S}[n+1] = 0.7039X[n]+0.2229X[n-1], \quad MSE = \sigma_\varepsilon^2 = 2.6687 > 2$$

Obviously, in the presence of AWGN the MSE increases.